Low-Rank Control Interfaces in Multiscale Competency Architectures

A Causal Criterion for Coordination Across Scales

Gunnar Zarncke

July 2026

Abstract

Biological systems coordinate competent subsystems across scales without specifying every lower-level action. Cells repair tissues, tissues restore anatomy, and organisms regulate physiology despite perturbations that alter the microscopic implementation. Michael Levin describes this organization as a *multiscale competency architecture*. We study one candidate mechanism of cross-scale composition: a restricted control interface through which few independent command directions preserve most of the viability obtainable by unrestricted lower-level control.

The proposal is deliberately narrower than the claim that biological systems compress information. We define a low-rank control interface by its contribution to finite-horizon viability under a specified disturbance family. Two quantities suffice: a normalized viability-retention curve $R_T(r;\Delta)$, comparing the best rank-r interface with null and unrestricted-control baselines, and the minimal viability-preserving rank $r_{\varepsilon,T}(\Delta)$. We connect these quantities to viability theory, feedback refinement relations, bisimulation, structural causal models, and low-rank approximation.

Three formal results carry the argument. First, bidirectional feedback refinement makes viability at a coarse scale equivalent to viability of the corresponding concrete states, giving a precise condition under which a higher-level controller can ignore lower-level detail. Second, for a locally linear disturbance-correction problem, the optimal rank-r interface is determined by the leading singular directions of the whitened interventional response, and the neglected singular-value tail yields an explicit upper bound on viability loss. Third, observational data cannot identify causal interface rank: structural causal models with identical observational distributions can have arbitrary different interventional ranks. We operationalize the framework for bacterial chemotaxis and planarian morphogenesis, and use muscle synergies as a negative control where observational low dimensionality does not establish a low-rank controller. The framework may eventually shed light on human values by separating the semantic complexity of learned value content from the dimensionality of the channels through which it changes policy.

1 Introduction

A multicellular organism contains an enormous number of degrees of freedom, yet successful regulation rarely resembles centralized micromanagement. A wound does not issue a complete cell-by-cell construction plan. A nervous system does not normally command each motor unit independently. A bacterial chemotaxis circuit does not represent every environmental molecule at the motor. Across such cases, lower-level components absorb variation while a smaller set of signals changes the larger-scale outcome.

This pattern is easy to overstate. Every interface discards detail, and almost every scientific model reduces dimension. It would therefore be nearly vacuous to say that biological organization relies on compression or bottlenecks. The substantive question is narrower:

Can a small number of independent intervention directions preserve a larger-scale competency across perturbations, relative to what could be achieved with unrestricted lower-level control?

This question is naturally situated inside Levin's multiscale competency architecture (MCA) framework Fields and Levin (2022); Levin (2023). MCA emphasizes that biological components at many scales exhibit regulative plasticity: they pursue preferred regions in metabolic, physiological, morphological, and behavioral problem spaces despite perturbations. The present paper does not attempt to replace or subsume that program. It isolates one possible mechanism by which competencies at adjacent scales compose.

The historical structural precursor is Simon's near decomposability Simon (1962). In a nearly decomposable system, within-module interactions dominate on short timescales, while slower aggregate variables mediate interactions among modules. Near decomposability explains why cross-scale variables may exist. It does not by itself show that a few such variables preserve adaptive function under intervention.

Friston's Markov-blanket framework supplies a related account of statistical boundaries Friston (2010); Kirchhoff et al. (2018). A blanket identifies variables through which internal and external states are conditionally coupled. A blanket may be large, and conditional independence does not imply that a small set of intervention directions controls the system's viable behavior. Boundaries and low-rank control therefore answer different questions.

Pearl's structural causal models (SCMs) supply the required distinction between observing a variable and changing it Pearl (2009, 2010). An observationally predictive latent may be a downstream readout, a common-cause proxy, or a mechanically induced correlation. A control interface must be evaluated under interventions. This point matters especially in motor-control and representation-learning settings, where low-dimensional observational structure is common but causally ambiguous.

The mathematical center of the paper is viability theory Aubin (1991); Aubin et al. (2011). A competency is represented operationally by a constraint or target set that the system can remain in, return to, or reach despite disturbances. This does not imply that every biological goal is literally a fixed set. It provides a standard and checkable approximation for a specified experiment.

The paper makes four contributions.

1. It defines a rank-dependent viability-retention curve with explicit null and unrestricted-control baselines.
2. It derives three results connecting multiscale control to feedback refinement, low-rank interventional response, and causal non-identifiability.
3. It gives a common experimental protocol for bacterial chemotaxis, planarian morphogenesis, and motor coordination.
4. It distinguishes the low rank of a control channel from low description length of the competencies or values implemented behind that channel.

The claim is conditional and falsifiable. Some multiscale competencies may require distributed, high-rank control. Others may use several context-dependent interfaces rather than one stable interface. A system can also appear low-dimensional only because the experiment explores a narrow task family. The proposed quantities are designed to expose these failures rather than redescribe them as successes.

2 Conceptual and Technical Positioning

2.1 Levin: multiscale competency architectures

Fields and Levin characterize cognition in terms of navigation through problem spaces toward preferred regions, allowing comparison across very different embodiments Fields and Levin

(2022). Levin further argues that development and regeneration exploit competent materials whose local capabilities can be harnessed by evolution rather than constructed from passive parts Levin (2023). The resulting MCA contains nested competencies: molecular networks regulate chemical states, cells regulate cellular states, tissues regulate anatomy, and organisms regulate behavior.

The central engineering advantage is *delegation*. A higher scale can specify a target or error without specifying the lower-scale trajectory that realizes it. Lower-level competencies fill in implementation detail, compensate for damage, and reuse existing physiological machinery.

We formalize this delegation with three standard objects:

MCA term	**Control-theoretic counterpart used here**
Preferred region or goal state	Target set, safe set, or terminal set K
Regulative plasticity	Viability or robust controlled invariance under a disturbance class
Cross-scale control interface	Restricted command space and a refinement map to lower-level controls
Lower-level competency	Controller that realizes abstract commands while absorbing within-fiber disturbances
Problem space	State or output space in which success constraints are defined

This translation is intentionally partial. MCA includes learning, goal plasticity, scale changes, and the formation of new problem spaces. The current paper treats the simpler question of preserving a specified competency over a specified horizon.

2.2 Simon: near decomposability

Simon observed that many complex systems are hierarchically organized and nearly decomposable: interactions within subsystems are stronger or faster than interactions between subsystems Simon (1962). After fast local transients decay, aggregate variables can describe slower cross-module dynamics. The proposal here can be viewed as an interventional strengthening of that observation.

Near decomposability gives a timescale and coupling condition under which a coarse state may be predictive. A low-rank control interface additionally requires that a restricted command space preserve the relevant viability. The implication is one-way:

$$\text{near decomposability} \not\Rightarrow \text{low-rank viability-preserving control},$$

but near decomposability can make such control possible by allowing local dynamics to settle inside fibers of a coarse description.

2.3 Friston: boundaries and sufficient statistics

Markov blankets partition variables into internal, external, sensory, and active sets so that internal and external states become conditionally independent given the blanket Friston (2010); Kirchhoff et al. (2018). This is useful for locating candidate boundaries and candidate channels. It does not determine whether the active states have low intervention rank, nor whether interventions through them preserve a larger-scale competency.

A blanket is therefore best treated as a search-space restriction. Once a boundary is identified, the present framework asks how the viable behavior of the bounded system changes when the available active-state interventions are restricted to a rank-r command family.

This distinction also clarifies the relation to unsupervised agent discovery (UAD), which searches raw dynamics for approximate blanket-like boundaries Zarncke (2025b). UAD can propose where an adaptive unit and its boundary might be. Low-rank control-interface discovery asks which interventions through that boundary preserve the unit's competency.

2.4 Pearl: intervention rather than association

For an SCM, the quantity

$$P(Y \mid Z = z)$$

describes an observational conditional, whereas

$$P(Y \mid \mathrm{do}(Z = z))$$

describes the distribution after replacing the structural equation for Z by the assignment $Z = z$ Pearl (2009, 2010). The difference is not cosmetic. A low-dimensional Z may predict Y because it is caused by Y, because both share a cause, or because task mechanics constrain them jointly. Only the interventional distribution can establish that Z is a control channel.

Pearl's machinery is used here at two levels. First, the response of a candidate interface is defined interventionally. Second, the non-identifiability theorem in Section 4.3 shows that observational data alone cannot identify causal interface rank, even with infinite samples and exact knowledge of the joint distribution.

3 Formal Setting

3.1 Controlled dynamics and viability

Let a discrete-time controlled system be

$$\Sigma = (X, U, W, F),$$

where X is the state space, U the admissible control set, W the disturbance set, and

$$x_{t+1} \in F(x_t, u_t, w_t).$$

Set-valued dynamics allow uncertainty, unmodeled lower-level detail, and nondeterministic responses. A deterministic system is the special case in which F is single-valued.

Let $K \subseteq X$ encode the competency-relevant constraint. For homeostasis, K may be a physiological safe range. For regeneration, it may be the set of trajectories that terminate in an admissible anatomy. For chemotaxis, it may be a position-energy region or a minimum nutrient-acquisition condition.

For worst-case disturbances, the finite-horizon viability kernel is

$$\mathrm{Viab}_T^{\Sigma}(K) = \left\{ x_0 \in K : \exists \pi,\ \forall w_{0:T-1} \in W^T,\ x_t \in K \text{ for all } 0 \le t \le T \right\}. \tag{1}$$

The infinite-horizon kernel is obtained by requiring the condition for all T. This is the standard viability-theoretic notion: the largest subset from which some admissible feedback can satisfy the constraints Aubin (1991); Aubin et al. (2011).

Biological experiments more often estimate a probability than a worst-case set. Let ρ_0 be a distribution over initial states and let ν_Δ be a disturbance distribution indexed by severity Δ. For a controller class Π, define

$$S_T(\Pi; \Delta) = \sup_{\pi \in \Pi} \mathbb{P}_{\rho_0, \nu_\Delta, \pi} \left[x_t \in K \text{ for all } 0 \le t \le T \right]. \tag{2}$$

This is a stochastic finite-horizon viability probability. The disturbance index Δ can be experimentally concrete: ligand noise amplitude, fraction of cells ablated, duration of gap-junction blockade, or external force magnitude.

3.2 Control interfaces

A rank-r interface consists of a command space $V_r \subseteq \mathbb{R}^r$ and a refinement map

$$\alpha_r : X \times V_r \to U.$$

A higher-level controller chooses $v_t \in V_r$; the refinement map converts it into lower-level actuation $u_t = \alpha_r(x_t, v_t)$. Dependence on x_t allows the lower level to implement the same abstract command differently in different microstates.

The rank is the dimension of the independent command space. For smooth nonlinear interfaces, this is replaced locally by the maximal differential rank of $v \mapsto \alpha_r(x, v)$ on the operating region. This uses standard input dimension rather than a new checklist of interface properties.

Let $\Pi_r(\alpha_r)$ be the set of concrete controllers implementable through interface α_r. We distinguish three baselines:

- Π_{full}: the unrestricted lower-level controller class;
- Π_{null}: a fixed, absent, scrambled, or ablated interface baseline;
- Π_r: the best allowed rank-r interface/controller class under the experimental protocol.

The phrase "best rank-r" is relative to a specified search space. In a synthetic model, all rank-r linear subspaces may be optimized. In a biological experiment, the search may be restricted to combinations of known channels, stimulation patterns, or independently addressable modes.

3.3 The viability-retention curve

Define

$$S_T^{\text{full}}(\Delta) = S_T(\Pi_{\text{full}}; \Delta), \qquad S_T^{\text{null}}(\Delta) = S_T(\Pi_{\text{null}}; \Delta),$$

and

$$S_T^{\star}(r; \Delta) = \sup_{\alpha_r} S_T(\Pi_r(\alpha_r); \Delta).$$

Assuming $S_T^{\text{full}}(\Delta) > S_T^{\text{null}}(\Delta)$, define the normalized viability retention

$$R_T(r; \Delta) = \frac{S_T^{\star}(r; \Delta) - S_T^{\text{null}}(\Delta)}{S_T^{\text{full}}(\Delta) - S_T^{\text{null}}(\Delta)}. \tag{3}$$

When the interface search space is nested in r, $R_T(r; \Delta)$ is nondecreasing and lies in $[0, 1]$. The normalization makes different systems comparable without pretending that a chemotactic success probability and a regeneration probability have the same natural unit.

The corresponding minimal viability-preserving rank is

$$r_{\varepsilon,T}(\Delta) = \min \left\{ r : R_T(r; \Delta) \geq 1 - \varepsilon \right\}. \tag{4}$$

Equations (3) and (4) are the only new measurement definitions required by the paper. Other quantities, such as recovery time or perturbation tolerance, enter by changing T or Δ and inspecting the same curve.

Remark 1. *A low value of $r_{\varepsilon,T}(\Delta)$ does not imply that the controlled competency has low description length. The refinement map α_r, the lower-level dynamics, and the target set may be extremely complex. A small command alphabet can select among complex, already learned or evolved behaviors.*

4 Three Formal Results

4.1 Result I: viability preservation under feedback refinement

Abstraction-based control studies when a controller synthesized on a coarse model can be refined into a controller for a detailed plant. Feedback refinement relations provide a standard sufficient condition Reissig et al. (2017); bisimulation provides a stronger two-way behavioral equivalence that preserves reachability and temporal properties Pappas (2003); Tabuada (2009). We state the viability consequence in a form adapted to MCA.

Let the concrete system be

$$\Sigma = (X, U, F)$$

and the abstract system

$$\bar{\Sigma} = (\bar{X}, \bar{U}, \bar{F}),$$

where uncertainty and disturbance have been absorbed into set-valued transitions. Let $Q \subseteq X \times \bar{X}$ relate concrete to abstract states. For a set $A \subseteq \bar{X}$, write

$$Q^{-1}(A) = \{x \in X : \exists \bar{x} \in A,\ (x, \bar{x}) \in Q\}.$$

A relation Q is a feedback refinement relation from Σ to $\bar{\Sigma}$ if every abstract input enabled at a related abstract state can be refined into a concrete input such that every concrete successor remains related to an allowed abstract successor. This is the control-compatible version of a simulation relation Reissig et al. (2017).

Proposition 1 (One-way viability preservation)**.** *Let Q be a feedback refinement relation from Σ to $\bar{\Sigma}$. Let $\bar{K} \subseteq \bar{X}$ and let $K = Q^{-1}(\bar{K})$. Then for every horizon T,*

$$Q^{-1}\Big(\mathrm{Viab}_T^{\bar{\Sigma}}(\bar{K})\Big) \subseteq \mathrm{Viab}_T^{\Sigma}(K). \tag{5}$$

If the inverse relation Q^{-1} is also a feedback refinement relation, then

$$Q^{-1}\Big(\mathrm{Viab}_T^{\bar{\Sigma}}(\bar{K})\Big) = \mathrm{Viab}_T^{\Sigma}(K). \tag{6}$$

Proof. Take $x_0 \in Q^{-1}(\mathrm{Viab}_T^{\bar{\Sigma}}(\bar{K}))$. There exists a related $\bar{x}_0 \in \mathrm{Viab}_T^{\bar{\Sigma}}(\bar{K})$ and an abstract feedback policy that keeps every abstract trajectory in $\bar{K}$ for T steps. By the feedback refinement property, each abstract action can be refined to a concrete action whose possible successors are related to possible abstract successors. Induction on t yields a concrete trajectory related at every step to an abstract trajectory in $\bar{K}$, hence contained in $K = Q^{-1}(\bar{K})$. This proves the inclusion. Applying the same argument to Q^{-1} gives the reverse inclusion under bidirectional refinement. □

Interpretation. Equation (6) is a precise version of cross-scale delegation. When the abstract and concrete dynamics are control-equivalent for the safety specification, a higher-level controller loses no viability by ignoring distinctions within the fibers of Q. The lower-level system can vary or repair itself inside a fiber without changing the higher-level policy.

The result connects three literatures without replacing them:

$$\begin{aligned}&\text{Simon's coarse aggregate} + \text{Levin's lower-level competency}\\ &\qquad + \text{feedback refinement/bisimulation} \Rightarrow \text{preserved viability.}\end{aligned}$$

One-way refinement is more realistic than exact equivalence. It certifies that an abstract controller is safe when refined, but it may be conservative: the concrete system can possess viable strategies absent from the abstraction. Empirically, the gap between the two sides of (5) measures how much lower-level competency the coarse interface fails to expose.

4.2 Result II: optimal local rank and a viability bound

The global computation of viability kernels is difficult. Around a target manifold or operating point, however, an intervention experiment often supplies a local response matrix. We derive a baseline connecting its singular spectrum to the best rank-r interface.

Suppose a disturbance creates a corrective demand $d \in \mathbb{R}^m$. An unrestricted controller could apply $u = d$. The competency-relevant output error after applying u is locally

$$e = G(d - u), \tag{7}$$

where $G \in \mathbb{R}^{q \times m}$ is the interventional response from residual lower-level error to the macro-outcome. Let d have mean zero and covariance $\Sigma \succ 0$. A rank-r linear interface applies

$$u = Pd,$$

where P is a rank-r projector in the whitened demand coordinates.

Let

$$M = G\Sigma^{1/2}$$

have singular values

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_s \geq 0.$$

Theorem 1 (Best rank-r local interface)**.** *Among rank-r orthogonal projections in whitened demand coordinates, the minimum expected squared macro-error is*

$$\min_{\text{rank}(P)=r} \mathbb{E}\|e\|_2^2 = \sum_{i>r} \sigma_i^2(M). \tag{8}$$

An optimum projects onto the leading r right singular vectors of M.

Proof. Write $d = \Sigma^{1/2} z$ with $\mathbb{E}[zz^\top] = I$. For a projector $\tilde{P}$ in whitened coordinates,

$$e = M(I - \tilde{P})z.$$

Therefore

$$\mathbb{E}\|e\|_2^2 = \|M(I - \tilde{P})\|_F^2.$$

The Eckart–Young–Mirsky theorem implies that the best rank-r approximation to M in Frobenius norm retains its leading r singular directions, leaving squared error equal to the tail sum in (8) Eckart and Young (1936); Mirsky (1960). □

Suppose local viability requires $\|e\|_2 < m$, where m is the distance to the nearest competency-violating boundary in the chosen output metric.

Corollary 1 (Finite-horizon one-step viability bound)**.** *For the optimal rank-r interface,*

$$\mathbb{P}(\|e\|_2 \geq m) \leq \frac{\sum_{i>r} \sigma_i^2(M)}{m^2}. \tag{9}$$

Hence

$$\mathbb{P}(\|e\|_2 < m) \geq 1 - \frac{\sum_{i>r} \sigma_i^2(M)}{m^2}. \tag{10}$$

Proof. Apply Markov's inequality to the nonnegative random variable $\|e\|_2^2$ and substitute Theorem 1. □

The bound is conservative, but it pays rent in three ways.

First, it gives an experimentally calculable baseline. Estimate G by randomized interventions, estimate Σ from the disturbance ensemble, and measure the local margin m. The singular-value tail then predicts how many independent interface directions should suffice.

Second, it distinguishes low-rank *control response* from low-dimensional observations. The SVD is applied to an interventional response matrix, not to passive trajectories.

Third, it yields a falsification test. If the empirical $R_T(r;\Delta)$ is much lower than the local bound predicts, then nonlinearities, state dependence, temporal effects, or unobserved bypasses matter. If it is much higher, lower-level feedback may be absorbing disturbances that the local open-loop response treats as residual error.

Relation to controllability and model reduction. The result is related to output controllability, controllability Gramians, balanced truncation, and input-output model reduction Moore (1981); Antoulas (2005). It is not equivalent to full-state controllability. A system can require many directions to reach arbitrary microstates while requiring few directions to keep a macro-output inside a viable set. The relevant spectrum is therefore the response of competency variables to interventions, not the rank of the full controllability matrix.

4.3 Result III: causal rank is not observationally identifiable

We now show that even exact knowledge of the observational distribution cannot determine whether a predictive low-dimensional variable is a control interface.

Theorem 2 (Observational non-identifiability of interface rank)**.** *For any matrix $A \in \mathbb{R}^{q\times k}$, there exist two SCMs with identical observational distributions over (Z, Y) but interventional response ranks* $\operatorname{rank}(A)$ *and* 0*, respectively.*

Proof. Let $H \in \mathbb{R}^k$ and $\varepsilon \in \mathbb{R}^q$ be independent random variables. Consider

$$\mathcal{M}_1: \qquad Z := H, \qquad Y := AZ + \varepsilon,$$

and

$$\mathcal{M}_2: \qquad Z := H, \qquad Y := AH + \varepsilon.$$

Observationally $Z = H$, so both models induce exactly the same joint distribution:

$$Y = AZ + \varepsilon.$$

Under intervention, however,

$$\mathbb{E}[Y \mid \mathrm{do}(Z = z)] = Az + \mathbb{E}[\varepsilon]$$

in $\mathcal{M}_1$, whose derivative with respect to z has rank $\operatorname{rank}(A)$. In $\mathcal{M}_2$, intervening on Z leaves H unchanged, so

$$\mathbb{E}[Y \mid \mathrm{do}(Z = z)] = A\mathbb{E}[H] + \mathbb{E}[\varepsilon],$$

which is independent of z and has rank 0. $\square$

Corollary 2. *No statistic computed solely from the observational joint distribution $P(Z, Y)$, including mutual information, predictive sufficiency, principal-component rank, or representation dimension, can identify the causal rank of $Z \to Y$ without additional causal assumptions or interventions.*

This result is elementary but consequential. It rules out a common shortcut: observing that many downstream variables covary with a low-dimensional latent does not establish that the system coordinates through that latent. The same data can arise when the latent is a passive readout of a hidden common cause.

Pearlian causal assumptions can sometimes identify intervention effects from observational data, for example through back-door or front-door criteria Pearl (2009). In the biological cases emphasized here, direct perturbation is often more credible because the candidate interface is experimentally manipulable. Where interventions are limited, causal identifiability assumptions should be stated as part of the result, not hidden inside a representation-learning method.

5 Measurement Protocol

The common empirical object is the curve

$$R_T(r; \Delta).$$

A measurement protocol should make each symbol concrete before data collection.

5.1 Step 1: specify the competency set and horizon

Choose K and T so that success is externally checkable. Examples include:

- remaining inside a nutrient-rich spatial region for T minutes;
- regenerating one anterior and one posterior pole by a fixed day;
- keeping endpoint force or gait stability inside a tolerance tube for T seconds.

A target should not be selected only because the candidate interface predicts it well. It should be justified independently as a competency-relevant outcome.

5.2 Step 2: define a disturbance family

Specify ν_Δ, including the intervention timing and a scalar severity index. The same system may be low-rank for one disturbance family and high-rank for another. This dependence is not a nuisance. It is part of the claim.

Useful disturbance families vary one axis at a time initially:

- sensory noise or gradient reversals;
- cell deletion, tissue cuts, or transient channel blockade;
- forces, loads, or actuator failures.

5.3 Step 3: establish null and full baselines

The null baseline should remove or scramble the candidate control channel while preserving as much unrelated function as possible. The full baseline should expose all experimentally available independent control directions, not an imagined omnipotent controller.

This makes the normalization in (3) empirically checkable. It also prevents a weak rank-r interface from looking impressive merely because the task is easy without control.

5.4 Step 4: construct nested interface families

For each r, construct a nested family of independently controllable modes. Depending on the system, these may be:

- biochemical concentrations or pulse modes;
- spatial voltage eigenmodes or optogenetic stimulation patterns;
- neural stimulation components or muscle-command modes.

Whenever possible, randomize interventions within the allowed family and estimate the interventional response matrix G. The leading singular directions from Theorem 1 provide a baseline ordering. A nonlinear controller can then be compared against this local linear benchmark.

5.5 Step 5: estimate retention and rank

Estimate S_T^{null}, $S_T^{\star}(r)$, and S_T^{full} with binomial or survival-model uncertainty intervals. Report the full $R_T(r;\Delta)$ curve rather than only $r_{\varepsilon,T}$. A sharp elbow supports a compact interface; a gradual curve indicates distributed control; crossings across Δ indicate context-dependent rank.

The primary comparisons are:

$$R_T(r;\Delta) \quad \text{against} \quad R_T(0;\Delta) = 0 \quad \text{and} \quad R_T(r_{\text{full}};\Delta) = 1,$$

and

$$r_{\varepsilon,T}(\Delta) \quad \text{across disturbance severities and systems.}$$

No additional case-specific score is necessary.

6 Canonical Cases

6.1 Bacterial chemotaxis: a minimal control-interface case

The *Escherichia coli* chemotaxis system maps temporally varying receptor occupancy into run-and-tumble behavior. Receptor methylation supports adaptation, the kinase pathway changes the concentration of phosphorylated CheY, and CheY-P binds the flagellar motor switch to alter clockwise versus counterclockwise rotation Clausznitzer et al. (2010); Yuan et al. (2012); Tu (2013). The pathway is attractive as a minimal case because the molecular components, perturbations, and motor output are experimentally accessible.

6.1.1 Candidate abstraction

A useful hierarchy is:

$$\begin{aligned}\text{ligand and receptor microstate} &\longrightarrow \text{CheY-P concentration} \longrightarrow \text{motor bias}\\ &\longrightarrow \text{spatial nutrient acquisition.}\end{aligned}$$

At the motor-switch level, CheY-P is approximately a scalar broadcast variable. This yields a concrete upper bound of one on the instantaneous command dimension for changing motor bias under a fixed receptor and motor context. It does *not* establish that the entire chemotactic competency has rank one. Receptor adaptation, multiple motors, motor-level adaptation, and cell-state dependence may add independent directions or state variables Yuan et al. (2012).

6.1.2 Common estimand

Let K be the set of trajectories that keep the cell above a specified nutrient-acquisition or energy threshold for T. Let Δ index gradient noise, reversal frequency, or background-concentration variation.

The baselines are experimentally direct:

- **Null**: CheY absent, nonphosphorylatable, clamped, or scrambled so that ligand changes no longer modulate motor bias.
- **Rank one**: a controlled CheY-P trajectory, or an optogenetic/chemical proxy that supplies one scalar time-varying command.

- **Full**: all independently accessible pathway interventions, including receptor, kinase, adaptation, and motor-level control.

The central measurement is

$$R_T(1;\Delta).$$

If it remains near one across a wide disturbance range, the scalar CheY-P channel preserves most experimentally available chemotactic viability. If retention falls under changing backgrounds while a higher-rank controller restores performance, the system is low-rank only locally.

6.1.3 Checkable baseline

The local response from CheY-P to motor bias is steep and experimentally measured, while motor adaptation shifts the response curve as the steady-state CheY-P level changes Yuan et al. (2012). This supplies a direct test of the local linear theorem: estimate the interventional response around several operating points and ask whether a single direction predicts viability retention. The known operating-point dependence is expected to break a globally fixed linear interface, providing a useful nontrivial baseline rather than a guaranteed success.

6.2 Planarian bioelectric morphogenesis: the central multiscale case

Planarian regeneration is a clearer instance of MCA because lower-level cells reconstruct large-scale anatomy after cutting. Bioelectric states generated by ion channels, pumps, and gap junctions participate in anterior-posterior patterning and anatomical memory Levin (2014); Durant et al. (2017). Transient perturbation of endogenous bioelectric networks can produce persistent changes in the anatomy to which later fragments regenerate.

Durant and colleagues reported that brief gap-junction perturbation produced a stable mixture of normal and two-headed regenerates; morphologically normal treated animals retained a cryptic altered regenerative state, and later amputations reproduced the altered outcome ratio even after the original treatment was absent Durant et al. (2017). Experimental reversal of the bioelectric state restored wild-type regeneration. This is strong evidence that a manipulable physiological state lies upstream of a complex anatomical outcome. It is not yet evidence that the full interface has low rank.

6.2.1 Candidate abstraction

Let x include cellular voltage, gap-junction state, gene expression, neoblast state, and local geometry. Let the macrostate $y = h(x)$ describe regenerative polarity and head number. The higher-level target set is

$$\bar{K}_{\mathrm{WT}} = \{\text{one anterior head and one posterior tail}\}.$$

The concrete target set is its preimage under h.

The MCA interpretation is that lower-level cellular competencies implement a coarse anatomical target despite large changes in cell identity and position. Proposition 1 gives the exact idealization: if anatomical states and cellular states are related by a feedback refinement relation, a controller defined over anatomy can be safely refined into cellular dynamics.

6.2.2 Common estimand

Let Δ index cut geometry, removed tissue fraction, pharmacological perturbation duration, or targeted cell ablation. Define success at horizon T as regeneration into the chosen target morphology.

Construct spatially controlled voltage modes, gap-junction perturbation modes, or combinations of ion-channel interventions. Then estimate

$$R_T(r; \Delta)$$

for nested mode families.

The baselines are:

- **Null**: sham stimulation or a perturbation that destroys the instructive spatial relation while matching exposure.
- **Rank** r: the best r independently controlled bioelectric modes.
- **Full**: the full experimentally addressable stimulation basis, possibly combined with known pharmacological channel controls.

The 2017 result already supplies a checkable one-direction intervention effect: a transient manipulation changes the long-run distribution over anatomical attractors, and a reverse intervention resets it Durant et al. (2017). It therefore establishes causal leverage and persistence. It does not provide $r_{\varepsilon,T}$, because the study did not compare nested control bases against a full-control baseline. The proposed experiment converts the qualitative "bioelectric code" claim into a rank-retention curve.

6.2.3 Decisive outcomes

Three patterns would discriminate hypotheses.

1. A sharp elbow at small r, stable across cut geometries, would support a low-rank anatomical control interface.
2. A low rank for simple anterior-posterior polarity but a rising rank for head shape, organ placement, and scaling would support task-relative hierarchy rather than a single anatomical code.
3. A high rank with no stable modes across perturbations would suggest distributed morphogenetic control, even though individual bioelectric interventions remain powerful.

The framework therefore supports Levin's program whether the low-rank hypothesis succeeds or fails: it measures how much anatomical competency is delegated through a compact interface rather than presuming that it is.

6.3 Muscle synergies: a negative control for forced mappings

Muscle activity often lies near a low-dimensional subspace, motivating the hypothesis that the nervous system controls groups of muscles through reusable synergies Bizzi and Cheung (2013). This appears, at first, to be an ideal example of a low-rank control interface.

Kutch and Valero-Cuevas showed why that inference is unsafe Kutch and Valero-Cuevas (2012). Musculoskeletal geometry and task constraints can produce low-dimensional electromyographic patterns even when muscles receive independent commands. The observed rank may therefore reflect the feasible output manifold rather than the controller's input dimension.

This case directly instantiates Theorem 2. Passive recordings cannot distinguish a neural low-rank controller from a high-rank controller acting through low-dimensional biomechanics.

6.3.1 Common estimand

Let K be a task-level tolerance tube for endpoint force, posture, or gait. Let Δ index external force or actuator impairment. Compare:

- **Null**: stimulation or command patterns uncorrelated with task-relevant modes;
- **Rank** r: direct stimulation of r independently calibrated neural or muscular command modes;
- **Full**: independently addressable stimulation of the available muscles or motor pools.

Table 1: Common quantities and evidence status.

Case	Competency set K	Candidate interface	Known causal fact	Missing rank test
E. coli chemotaxis	Nutrient acquisition or spatial retention	CheY-P and pathway modes	CheY-P intervention changes motor bias; adaptation changes operating point	Nested pathway modes versus full accessible control
Planarian regeneration	Target morphology after cut	Spatial bioelectric and gap-junction modes	Transient perturbation persistently changes anatomical outcome distribution; reversal can reset	Nested spatial modes across cut and damage families
Motor coordination	Force, posture, or gait tolerance	Neural or muscle-command modes	Stimulation can evoke structured patterns, but passive low rank has mechanical alternatives	Direct rank-r command interventions versus independent-control baseline

Only interventional estimates of

$$R_T(r;\Delta)$$

can establish whether few command modes preserve task viability. PCA or nonnegative matrix factorization of EMG can propose candidate modes but cannot supply the causal conclusion.

As a negative control, muscle coordination sets a demanding standard for the other cases. A candidate interface should not be accepted merely because a low-dimensional factor predicts the macro-outcome. The same nested intervention protocol must outperform both null and mechanically induced observational baselines.

7 Cross-Case Comparison

Table 1 summarizes what is currently established and what remains to be measured. The table intentionally avoids assigning speculative numerical ranks.

The same two reported objects apply to all three:

$$R_T(r;\Delta) \qquad \text{and} \qquad r_{\varepsilon,T}(\Delta).$$

This provides three useful comparisons.

Within-system comparison. Does the required rank increase with perturbation severity? A control interface that appears scalar in a narrow operating regime may recruit additional modes under damage.

Across-task comparison. Does the same interface preserve multiple competencies? Shared low-rank control predicts correlated failures and common-mode leverage; task-specific interfaces predict different rank curves.

Against an observational baseline. How does interventional rank compare with the rank obtained from PCA, factor analysis, or latent-state models? A lower observational rank than interventional rank indicates compression without control. A lower interventional rank than observational rank indicates that many observed states can be steered through few commands because lower-level feedback supplies the detail.

8 Why Low-Rank Interfaces Might Emerge

The framework defines and measures a property. It does not by itself explain why the property should be common. Several mechanisms make it plausible.

8.1 Timescale separation

In nearly decomposable systems, fast local dynamics relax before slow cross-module dynamics change substantially Simon (1962). A higher-level controller can then act on aggregate variables while local controllers stabilize the omitted dimensions. Formally, timescale separation makes approximate feedback refinement more plausible because many microstates inside a coarse fiber have similar future coarse behavior.

8.2 Evolvability and reuse

If evolution can alter a small command interface while reusing competent lower-level machinery, large phenotypic changes require fewer coordinated genetic modifications. This does not mean that the resulting phenotype is simple. It means that the mapping from high-level signal to lower-level response has already been built and can be reused Levin (2023).

8.3 Communication and energetic constraints

Long-range broadcast is costly. Compact control signals can coordinate distributed subsystems without transmitting their full local states. Neuromodulators, hormones, and bioelectric fields are plausible implementations, but the physical narrowness of a broadcast channel does not guarantee low $r_{\varepsilon,T}$. Context-dependent receptor states can effectively increase the dimension of the interface.

8.4 Credit assignment

A shared control channel can carry outcome-relevant error or precision signals to many local learning systems. This may accelerate adaptation while leaving semantic content distributed. The loop-hub-value proposal applies a related idea to neuromodulatory control of human motivation Zarncke (2025a). The present framework supplies a way to test the architectural part of that proposal without assuming a one-to-one mapping between hubs and value concepts.

8.5 Why high-rank control may persist

Low rank also creates common-mode vulnerabilities. A pathogen, tumor, or adversarial intervention that captures a central interface can redirect many downstream processes at once. Redundant or distributed control can be selected precisely because it is harder to capture. The empirical prediction is not that rank is always small, but that rank reflects a tradeoff among evolvability, communication cost, robustness to ordinary variation, and robustness to interface corruption.

9 Applications Beyond the Canonical Cases

9.1 Motivational control and human values

Human values are semantically rich, learned, culturally shaped, and internally inconsistent. None of this determines the dimensionality of the control channels through which motivational state changes policy.

Suppose a high-dimensional learned world model and concept system is modulated by a smaller family of affective, neuromodulatory, or self-referential control signals. Then value *content* may have high description length even if policy changes lie on a low-rank intervention manifold. Conversely, finding a small preference direction in a trained model does not show that the represented concept of goodness is simple; the direction may rely on distributed semantic machinery.

The relevant experiment is therefore not PCA over value judgments alone. It is a rank-retention experiment:

1. specify a family of value-relevant policy competencies;
2. perturb candidate motivational channels in independently controlled combinations;
3. measure whether a small r reproduces the policy changes obtainable through the full accessible intervention set;
4. test stability across contexts, development, reflection, and distribution shift.

This will not solve value learning. It may shed light on human values by separating two questions that are often conflated:

$$\text{How complex is what humans value?} \qquad \text{versus} \qquad \text{How many control directions change policy?}$$

A low value of $r_{\varepsilon,T}$ would make motivational control more identifiable and manipulable, but also more vulnerable to common-mode capture. A high value would weaken simple hub-based value models and make stable value inference harder. The low-dimensional value-learning argument should therefore be treated as an application hypothesis rather than evidence for the general framework Zarncke (2026).

9.2 Conscious access and self-models

Global-workspace and metacognitive theories propose that only a restricted subset of neural states becomes jointly available for report, planning, and reflection. A limited bandwidth is not by itself a low-rank control interface. The stronger claim would be that interventions on a small set of workspace or self-model directions reproduce most changes in globally coordinated policy.

The same retention protocol can be used, with competency sets defined by cross-modal report, sequential planning, or ownership attribution. These applications require ethically and technically difficult interventions, so they are secondary to the biological systems above.

9.3 AI correction channels

AI systems are often trained through architecturally narrow feedback channels: scalar rewards, preference labels, constitutions, critique messages, or gradient updates. Their nominal interface dimension is easy to count, but nominal narrowness does not imply continued control. A capable system may learn internal routes that make behavior insensitive to the correction channel.

For an AI system, define K as a set of correction-responsive policy constraints and Δ as capability growth or distribution shift. Then

$$R_T(r;\Delta)$$

measures how much of the policy correction obtainable through the full training or editing stack remains available through a restricted channel. A decline with Δ would indicate loss of cross-scale control even if the interface still exists syntactically.

10 Failure Modes and Scope Conditions

10.1 Task-relative rank

The rank is always indexed by K, T, and Δ. A scalar interface may preserve one coarse competency while destroying another. Calling a system "low-rank" without these indices is incomplete.

10.2 Nonlinearity and mode switching

A nonlinear system may use different low-rank interfaces in different regions, while the union has high rank. Reporting only a local Jacobian spectrum can conceal switching. The empirical retention curve should therefore be measured across operating points.

10.3 Redundancy

Multiple interfaces may each preserve viability. Ablating one can leave performance unchanged, not because it is irrelevant but because another channel compensates. The optimization in $S_T^\star(r;\Delta)$ naturally allows alternative interfaces, but causal attribution to a particular biological channel requires combinatorial interventions.

10.4 Changing target sets

Development and learning can alter the competency itself. Viability theory assumes a specified constraint set. For target plasticity, one must enlarge the state to include target parameters or compare rank curves before and after target change.

10.5 Intervention damage

Manipulating a candidate interface may damage lower-level machinery, violating the assumption that only the command variable changed. Sham interventions and multiple implementation methods are necessary. Pearl's do-operator describes an ideal intervention; biological interventions only approximate it.

10.6 Search-space dependence

The "best rank-r" interface can be found exactly only in small synthetic systems or restricted linear families. Biological studies estimate a candidate-relative rank. The paper's notation should not be read as claiming global optimization when the experimental basis is limited.

10.7 Adversarial adaptation

A system can respond to the measurement process or route control around monitored channels. This is especially relevant for AI and social systems. Rank measured under a passive disturbance family may fail under strategic pressure.

11 A Minimal Synthetic Benchmark

Before applying the protocol to biological data, a benchmark should separate three properties that passive representation learning often conflates:

1. observational dimension;
2. causal response rank;
3. viability-preserving rank.

Consider a 20-dimensional microstate

$$x_{t+1} = Ax_t + Bu_t + Ew_t$$

with a two-dimensional macrostate

$$y_t = Cx_t$$

and competency set

$$K = \{x : \|Cx\|_\infty \leq 1\}.$$

Choose A so that 18 fiber directions are stable and rapidly contracting, while the two macro directions are unstable without control. Let B have full state-controllability rank 20, but let only two singular directions of the interventional map to y have appreciable gain.

The benchmark has an analytic expectation:

- full-state controllability remains high-rank;
- the best viability-preserving interface has $r_{\varepsilon,T} = 2$ over the designed disturbance range;
- PCA rank can be made arbitrarily large by adding high-variance stable fiber noise;
- a common-cause latent can be added to create a low-dimensional observational predictor with zero interventional rank, as in Theorem 2.

Four methods should be compared:

1. PCA or nonlinear latent dimension;
2. information-bottleneck prediction;
3. Markov-blanket or UAD boundary discovery;
4. randomized interventional estimation of G, followed by rank-r viability evaluation.

The intended result is not that intervention always wins. The benchmark identifies which question each method answers. PCA describes variance; information bottlenecks describe predictive compression; blanket methods describe conditional boundaries; the present method estimates restricted causal control of viability.

12 Research Program

A focused program can proceed in three stages.

12.1 Stage I: formal and synthetic

1. Generalize Theorem 1 from one-step linear response to finite-horizon stochastic systems using output-reachability Gramians.
2. Quantify approximate rather than exact feedback refinement and derive bounds on viability-kernel mismatch.
3. Release synthetic benchmarks where observational rank, causal rank, and viability rank vary independently.

12.2 Stage II: tractable biological systems

1. Estimate $R_T(1; \Delta)$ for CheY-P control across chemotactic operating regimes.
2. Construct nested spatial bioelectric stimulation bases for planarian regeneration.

3. Use motor-control experiments as a negative-control domain in which mechanical and neural low rank must be separated.

12.3 Stage III: cognition and alignment

Only after the measurement protocol succeeds on systems with known interventions should it be applied to motivational control, conscious access, self-models, and AI correction. These domains have richer semantics and worse causal access. Their importance is not a reason to relax the evidential standard.

13 Discussion

The paper began with the concern that "low-dimensional bottleneck" may merely restate compression and interfaces. The viability formulation identifies what would make the claim substantive.

A low-rank control interface is not just a small representation. It is a restricted command space that preserves most of the competency available through a richer controller, under a stated disturbance family. The comparison against null and full baselines makes the claim quantitative. The dependence on K, T, and Δ prevents a local or task-specific result from silently becoming a universal one.

The conceptual stack is deliberately asymmetrical.

- **Levin** supplies the target phenomenon: competencies compose across biological scales.
- **Simon** supplies the structural reason coarse interfaces can exist: near decomposability and timescale separation.
- **Friston** supplies a related method for locating statistical boundaries through which coupling occurs.
- **Pearl** supplies the distinction between predictive summaries and causes.
- **Aubin and control-abstraction theory** supply the formal criterion: preservation of viability under restricted and refined control.

This positioning is supportive rather than competitive. The framework does not claim that MCA is "really" viability theory, that Markov blankets are low-rank interfaces, or that all biological intelligence reduces to a single control principle. It offers a measurable subquestion within the broader agenda.

The strongest current biological evidence concerns causal leverage, not low rank. Chemotaxis contains a scalar motor-control signal, but its global competency is context-dependent. Bioelectric interventions can rewrite regenerative anatomy, but the number of independent modes required for near-complete control is unknown. Muscle synergies show why passive low-dimensionality cannot settle the issue.

The central empirical unknown is therefore not whether compact signals exist. They plainly do. It is the shape of

$$R_T(r; \Delta).$$

A steep curve would show that lower-level competencies absorb most variation behind a compact interface. A shallow curve would show that control remains distributed. A curve that changes sharply with Δ would show that low rank is a feature of ordinary operation but not of repair under severe damage.

This distinction also changes the alignment interpretation. A low-rank motivational or correction interface could make value influence easier to infer and reproduce. It would simultaneously create a concentrated failure mode. An interface can be a handle for alignment and an attack surface for misalignment. The framework measures the concentration before deciding whether it is desirable.

14 Conclusion

Multiscale competency architectures pose a concrete composition problem: how can a larger-scale system preserve a target while delegating implementation to competent lower-level parts? We proposed that some such systems use low-rank control interfaces, defined not by generic compression but by retained viability under restricted intervention.

The paper introduced one retention curve,

$$R_T(r;\Delta),$$

and one derived quantity,

$$r_{\varepsilon,T}(\Delta).$$

It connected exact preservation of viability to feedback refinement and bisimulation, derived a singular-value baseline and viability-loss bound for local linear response, and proved that causal interface rank is not identifiable from observations alone.

Bacterial chemotaxis and planarian regeneration provide strong candidate systems. Muscle synergies provide a warning against forced mappings. The next step is not another broad analogy. It is to estimate nested rank-retention curves under controlled perturbations. If small ranks preserve viability across cases and disturbance regimes, low-rank control interfaces will constitute a genuine recurring mechanism of multiscale competency. If they do not, the same experiments will identify where biological coordination remains irreducibly distributed.

A Deterministic and Stochastic Viability

For deterministic worst-case analysis, the success indicator in (2) can be replaced by the robust condition

$$\inf_{w_{0:T-1}\in W^T} \mathbf{1}\{x_t \in K,\ 0 \le t \le T\}.$$

Then S_T is either zero or one for a fixed initial state, and the useful aggregate is the probability mass or volume of the viability kernel:

$$\mathcal{V}_T(\Pi) = \rho_0\Big(\mathrm{Viab}_T^{\Pi}(K)\Big).$$

The same null/full normalization gives

$$R_T(r) = \frac{\mathcal{V}_T(\Pi_r) - \mathcal{V}_T(\Pi_{\text{null}})}{\mathcal{V}_T(\Pi_{\text{full}}) - \mathcal{V}_T(\Pi_{\text{null}})}.$$

This version is appropriate for gridded synthetic systems and formal reachability tools.

B Approximate Refinement

Exact feedback refinement is strong. Let an abstract safety controller keep abstract trajectories at least distance m from the boundary of $\bar{K}$. Suppose the concrete-to-abstract simulation error is bounded by $\delta < m$ in the output metric over horizon T. Then the refined concrete controller remains inside the δ-expansion of the abstract trajectory and therefore inside $\bar{K}$. This standard robustness-margin argument converts approximate bisimulation or approximate refinement bounds into safety certificates Tabuada (2009).

For MCA applications, the quantity of interest is not only the approximation error but its dependence on perturbation severity:

$$\delta = \delta(\Delta).$$

A lower-level competency is effective when $\delta(\Delta)$ remains below the macro-level margin without requiring an increase in command rank.

C Estimating the Interventional Response Matrix

Let $z \in \mathbb{R}^r$ parameterize a candidate stimulation basis and let $y \in \mathbb{R}^q$ be a vector of competency-relevant outcomes. Around operating point (z_0, c), define

$$G(c) = \frac{\partial}{\partial z} \mathbb{E}\left[Y \mid \mathrm{do}(Z = z), C = c\right]\Big|_{z=z_0}.$$

A randomized local design estimates columns of G by finite differences. For nonlinear systems, repeat across contexts c, operating points, and disturbance levels.

The singular values of $G(c)\Sigma^{1/2}(c)$ should be reported with bootstrap intervals. A stable low-rank interface predicts that leading singular subspaces align across contexts. Principal angles between the estimated subspaces provide a standard comparison; no new stability score is required.

D Relation to Information Bottlenecks

The information-bottleneck method seeks a representation Z that compresses X while preserving information about Y, often through an objective of the form

$$I(X;Z) - \beta I(Z;Y)$$

Tishby et al. (2000). This is an observational or predictive criterion unless the data and graphical assumptions identify interventions.

A control interface can be information-rich but low-rank, for example when a scalar time series has high temporal precision. It can also be information-poor but high-rank if many binary actuators each provide one independent direction. The relation between information rate and control rank is therefore empirical rather than definitional.

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